

Shape-Constrained Inference

Directed Reading Program — Final Report

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1 Introduction

A common setting in statistics is one where we know something qualitative about the function we are trying to estimate before we even look at the data. Maybe the relationship between age and performance is expected to go up before it comes down, or maybe the probability of a disease should only increase as a biomarker rises. These kinds of prior beliefs are called *shape constraints*, and the goal of shape-constrained inference is to build estimators that respect them without throwing out any flexibility we do not need to give up.

This report covers three interconnected ideas I worked through during the Directed Reading Program this quarter. The first is the **Pool Adjacent Violators Algorithm (PAVA)**, a simple and elegant algorithm for fitting a monotone non-decreasing regression to data. The second is the **Grenander estimator**, which uses the same isotonic regression machinery to produce a monotone-constrained density estimate. The third is **I-splines**, which generalize the approach by letting us enforce monotonicity through a flexible spline basis inside a regression model.

Across all three topics, the same theme comes up: when the shape constraint is actually correct, we get better estimates than we would from an unconstrained approach. And when we apply these methods to real data, the constraint often acts as a regularizer, smoothing out noise and making the fit easier to interpret.

2 Isotonic Regression and PAVA

2.1 The Isotonic Regression Problem

Suppose we observe data $(x_1, y_1), \dots, (x_n, y_n)$ where the x_i are already ordered, $x_1 \leq x_2 \leq \dots \leq x_n$, and we want to fit a regression function f such that $f(x_1) \leq f(x_2) \leq \dots \leq f(x_n)$ — that is, f is non-decreasing. This is the isotonic regression problem. Formally, we want to solve

$$\hat{f} = \arg \min_{f: f(x_1) \leq \dots \leq f(x_n)} \sum_{i=1}^n (y_i - f(x_i))^2.$$

The solution to this constrained least-squares problem has a beautiful closed form: it is a piecewise constant function where each piece is the average of consecutive groups of observations. The algorithm that computes it efficiently is PAVA.

2.2 The Pool Adjacent Violators Algorithm

The idea behind PAVA is straightforward. Start by treating each observation as its own block. Then scan left to right. Whenever you find a pair of adjacent blocks where the right block's average is *less* than the left block's average — a violation of monotonicity — you merge them and replace both with their weighted average. You keep scanning until no violations remain. Because merging can create new violations with earlier blocks, you may need to backtrack, which is why the algorithm is sometimes described as a *backward* pass after each merge.

More precisely, the algorithm maintains a stack of blocks. For each new observation y_i :

1. Push y_i as a new singleton block.
2. While the last two blocks violate the ordering, pop them both, merge into a new block with average equal to the weighted mean, and push the merged block.

The output is the sequence of block means, extended so that all observations within a block receive the same fitted value. The algorithm runs in $O(n)$ time, which makes it practical even for large datasets.

2.3 Statistical Properties

The isotonic regression estimator $\hat{f}$ is the maximum likelihood estimator under a Gaussian noise model subject to the monotone constraint. It is also consistent: as $n \rightarrow \infty$, $\hat{f}(x)$ converges to the true $f(x)$ at each point where f is strictly increasing. At a point where the true function is flat, the estimator is still consistent but converges more slowly.

The pointwise rate of convergence at a point x_0 where the true function has a nonzero derivative $f'(x_0) > 0$ is

$$\hat{f}(x_0) - f(x_0) = O_p(n^{-1/3}),$$

which is slower than the usual $n^{-1/2}$ parametric rate. This cube-root rate is a signature of shape-constrained estimators and shows up in multiple places throughout the theory.

The asymptotic distribution of $\hat{f}(x_0)$ at such a point is *not* Gaussian. Instead, after appropriate centering and scaling, it converges to the *Chernoff distribution*, which is defined as the slope at zero of the greatest convex minorant of a two-sided Brownian motion with a parabolic drift. The Chernoff distribution is what you use to build confidence intervals for the isotonic regression estimate — this is precisely how the confidence interval in the NBA application below was constructed.

2.4 Application: NBA Shooting Efficiency Across Age

To see PAVA in action I applied it to a real dataset of NBA player seasons from Basketball-Reference, focusing on ages 18 to 30, which covers the main development window of a professional basketball player’s career. The outcome variable is *true shooting percentage* (TS%), defined as

$$\text{TS\%} = \frac{\text{PTS}}{2(\text{FGA} + 0.44 \cdot \text{FTA})},$$

a composite efficiency metric that accounts for field goals, three-pointers, and free throws on a common scale.

The idea was that player efficiency should be monotone non-decreasing over this window — players generally get better at converting their shot attempts as they gain experience and refine their shot selection. I fit PAVA to the age-level means (rather than raw player-season observations) to avoid the algorithm being driven by the large variance in individual seasons.

The isotonic fit confirmed the expected upward trend, with the largest gains occurring during a player’s early 20s and the curve flattening out around age 26–28. I also fit a LOESS smoother for comparison. The LOESS curve tracks more local variation and actually dips slightly in the late 20s, whereas the isotonic fit is forced to remain non-decreasing. Whether the dip is real signal or noise is hard to say, but the isotonic version gives a cleaner and more interpretable summary of the overall trend.

For the point at age 28, the isotonic estimate was approximately 0.563, and I constructed a 95% confidence interval using the Chernoff distribution, yielding the interval [0.549, 0.578]. The Chernoff quantile replaces the usual normal quantile $z_{0.975} = 1.96$ because the asymptotic distribution is not Gaussian, and ignoring this would produce intervals with incorrect coverage.

I also extended the analysis to a bivariate surface, modeling the joint effect of age and minutes played (MP) on average points scored. Bivariate isotonic regression enforces monotonicity in both directions simultaneously — more minutes and older age should each increase expected output, at least within the development window. I approximated the bivariate solution using successive one-dimensional isotonic sweeps: first apply PAVA across the age dimension for each fixed MP level, then apply PAVA across the MP dimension for each fixed age. This iterative approach produces a surface that is monotone in both directions and can be visualized as a 3D plot.

3 The Grenander Estimator

3.1 Motivation

Isotonic regression constrains a regression function to be monotone. A natural question is whether we can do the same thing for density estimation: what if we know the underlying density should be non-increasing (or non-decreasing)? For example, waiting times between events, survival times, or empirical distributions of certain financial quantities are often modeled as having a decreasing density on $[0, \infty)$.

The *Grenander estimator*, introduced by Ulf Grenander in 1956, answers this question. It is the nonparametric maximum likelihood estimator of a monotone decreasing density, and it has a beautiful connection to PAVA through the concept of the greatest convex minorant.

3.2 Definition via the Empirical CDF

Let $X_1, \dots, X_n$ be i.i.d. observations from a density f on $[0, \infty)$ that is assumed to be non-increasing. The empirical CDF is

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq x).$$

The Grenander estimator $\hat{f}_n$ is defined as the left derivative of the least concave majorant (LCM) of $\hat{F}_n$. The LCM of $\hat{F}_n$ is the smallest concave function that lies everywhere above $\hat{F}_n$, and its left derivative gives a non-increasing step function — which is exactly the non-increasing density estimate we want.

Equivalently, $\hat{f}_n$ is piecewise constant and equals the slope of the LCM on each flat segment. The connection to PAVA is that computing the LCM of the empirical CDF is equivalent to applying an isotonic regression (in the direction of non-increasing rather than non-decreasing) to the slopes of the empirical CDF.

3.3 Statistical Properties

Like the isotonic regression estimator, the Grenander estimator converges at the cube-root rate $n^{-1/3}$ at any point x_0 where the true density $f(x_0) > 0$ and $f'(x_0) < 0$. The pointwise asymptotic distribution is again Chernoff, specifically

$$n^{1/3} \cdot \frac{\hat{f}_n(x_0) - f(x_0)}{c(x_0)} \xrightarrow{d} Z_{\text{Chernoff}},$$

where the scaling constant $c(x_0)$ depends on $f(x_0)$ and $|f'(x_0)|$.

One important property of the Grenander estimator is that it is *adaptive*: near a point where the true density has a steeper slope, the local blocks tend to be smaller (the PAVA-like merging stops sooner), so the estimator tracks the truth more closely. Near a flat region of the density, the blocks are wider and the estimate is smoother.

Another key result is that the Grenander estimator is a *consistent* estimator of the mode of the distribution, since the mode of a non-increasing density on $[0, \infty)$ is always at 0, and the estimator places the largest jump there.

3.4 Connection to PAVA

To see why the Grenander estimator is essentially PAVA in disguise, think about what the empirical CDF looks like between order statistics. Between $X_{(i)}$ and $X_{(i+1)}$, the empirical CDF is flat. So its “derivative” (loosely speaking) is zero there, while it has jumps of size $1/n$ at each order statistic. The LCM smooths these out by averaging the slopes over adjacent intervals, which is exactly what PAVA does when applied to slope values. This is why the Grenander estimator can be computed efficiently using the same underlying algorithm as isotonic regression, just applied to the increments of $\hat{F}_n$ rather than to observed y values.

3.5 Practical Considerations

In practice, the Grenander estimator can be implemented in R using the `fdrtool` or `logcondens` packages, or by computing the LCM of the ECDF manually using the `isotone` package. One practical issue is boundary behavior: at the left boundary, the estimate can be quite sensitive to the largest observed value, especially in small samples. The estimator also does not integrate to exactly one in finite samples unless we normalize it, since the LCM may not hit exactly 1 at the right edge of the observed data.

Despite these edge cases, the Grenander estimator is a powerful and theoretically grounded tool for density estimation under a monotonicity constraint. It requires no bandwidth selection (unlike kernel density estimation), and the monotonicity constraint serves as a built-in regularizer that prevents overfitting in tails where data are sparse.

4 I-Splines and Monotone Regression

4.1 Motivation

PAVA and the Grenander estimator both produce piecewise constant fitted values. This is fine for many purposes, but in regression settings we often want something smoother. I-splines provide a way to enforce monotonicity while also allowing a smooth, flexible fit.

The key idea is to build a basis for the space of monotone increasing functions. If every basis function is itself monotone increasing, then any non-negative linear combination of them is also monotone increasing. Fitting a regression model using this basis with the constraint that all coefficients are non-negative therefore gives a smooth, monotone estimate by construction.

4.2 M-Splines and Their Integrals

The starting point is *M-splines*, which are a normalized version of B-splines that integrate to one on their support (so they behave like local probability densities). Given a knot sequence $t_1 < t_2 < \dots < t_k$ and a degree d , the M-spline basis functions $M_1(x), \dots, M_m(x)$ are non-negative, have local support, and integrate to one.

I-splines are the integrals of M-splines:

$$I_j(x) = \int_{-\infty}^x M_j(t) dt.$$

Because $M_j \geq 0$ everywhere, each I_j is non-decreasing. Furthermore, $I_j(x) = 0$ for x below the support of M_j and $I_j(x) = 1$ for x above the support. This means each I-spline is a smooth, monotone increasing “switch” that turns on gradually as x increases through the j -th knot interval.

4.3 Constrained Regression with I-Splines

A monotone non-decreasing smooth function can be represented as

$$f(x) = \beta_0 + \sum_{j=1}^m \beta_j I_j(x), \quad \beta_j \geq 0 \text{ for } j = 1, \dots, m.$$

The non-negativity constraints on the β_j are what enforce monotonicity. If we allow $\beta_j < 0$, the sum could decrease somewhere.

Fitting this model is a constrained optimization problem: minimize the residual sum of

squares (or the negative log-likelihood for generalized linear models) subject to $\beta_j \geq 0$. This can be done efficiently using projected gradient methods or the L-BFGS-B algorithm, which handles box constraints.

4.4 Application: Predicting CKD from Serum Creatinine

I applied I-splines to predict the probability of chronic kidney disease (CKD) from serum creatinine levels, using a publicly available dataset. The clinical motivation for a monotone constraint is clear: higher serum creatinine indicates worse kidney function, so the probability of CKD should be non-decreasing in serum creatinine.

I built an I-spline basis with knots at the 25th, 50th, and 75th percentiles of the observed serum creatinine values and used degree 3, which gives smooth cubic pieces. The I-spline basis was then entered as predictors in a logistic regression model, with the constraint that all spline coefficients are non-negative. I fit the model using `optim` with the L-BFGS-B method in R, setting the lower bound to $-\infty$ for the intercept and 0 for all spline coefficients.

For comparison, I also fit an unconstrained logistic GLM (linear in serum creatinine) and a shape-constrained additive model (SCAM) using a monotone increasing P-spline smooth via the `scam` package. All three fits were plotted on the same axes to make the differences clear.

The unconstrained GLM is monotone by construction in this case (the coefficient on serum creatinine turned out to be positive), but it is restricted to a linear logit scale, which may miss nonlinearity in the relationship. The SCAM and I-spline fits both capture the nonlinear S-shape of the probability curve, rising more steeply in the middle range of creatinine values and flattening out at the extremes. The two constrained fits were very close to each other, which makes sense since they are estimating the same shape-constrained object using slightly different machinery.

The fitted I-spline coefficients were all non-negative at convergence, confirming that the constraint was active and not just redundant. This is important: it means the constraint is actually doing something, not just being satisfied automatically by the data.

4.5 Advantages over Piecewise Constant Methods

Compared to PAVA, I-splines offer two main advantages. First, the fit is smooth rather than piecewise constant, which is often more realistic for biological or physical relationships that vary continuously. Second, I-splines generalize naturally to the GLM setting (logistic regression, Poisson regression, etc.), whereas PAVA is fundamentally a least-squares method.

The tradeoff is that I-splines require choosing knot locations and the number of knots, whereas PAVA is completely nonparametric and requires no such choices.

5 Connecting the Three Methods

Looking back across all three methods, the common thread is that each one takes a qualitative belief about the shape of a function and bakes it directly into the estimator. PAVA does this in the simplest way possible, producing a piecewise constant monotone fit via a greedy averaging algorithm. The Grenander estimator does the same thing for density estimation, using the least concave majorant of the empirical CDF in a way that is mathematically equivalent to applying PAVA to the slopes. I-splines extend the idea to smooth regression and GLMs, giving up some of PAVA’s theoretical simplicity in exchange for a smoother and more flexible fit.

All three estimators share the same $n^{-1/3}$ convergence rate and Chernoff limiting distribution at interior points where the true function has a nonzero derivative. This is a universal feature of isotonic-type estimators and is one of the things that makes shape-constrained inference theoretically interesting: the rates and distributions are qualitatively different from the $n^{-1/2}$ Gaussian world of classical parametric statistics.

One practical lesson from working through these methods is that the monotonicity constraint acts as a regularizer in finite samples. When the data are noisy, the constraint prevents the estimator from chasing local fluctuations that are probably just noise. This was visible in the NBA application: the raw data had a lot of scatter around any smooth trend, and the PAVA fit provided a clean, interpretable summary of the overall age-efficiency relationship without overfitting the noise.

6 Conclusion

Shape-constrained inference is a rich area of nonparametric statistics that provides a middle ground between fully parametric models (too rigid) and unconstrained nonparametric methods (too flexible when we actually do have prior knowledge about the shape). The three methods covered in this report — PAVA, the Grenander estimator, and I-splines — each offer a different way to operationalize a monotonicity constraint, and each has strengths and weaknesses depending on the problem.

Going through these methods gave me a better sense of how prior structural knowledge can be formalized mathematically and built into an estimator. The applications to NBA shooting

efficiency and kidney disease prediction helped make the theory concrete: in both cases, the monotone constraint reflected real domain knowledge, and the constrained estimates were more interpretable and arguably more scientifically honest than unconstrained alternatives that might fit the observed data better but would violate what we actually know about the underlying process.

The DRP experience overall pushed me to engage with theory more carefully than I typically do in coursework. Reading through proofs of convergence rates and asymptotic distributions, rather than just accepting the results, gave me a much better intuition for why these methods behave the way they do — and for what kinds of problems they are (and are not) well-suited to handle.

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